Our Computational Tools

Simplex Alg – implementation via tableau.

specialized alg – Ford – Fulkerson

"not enough"

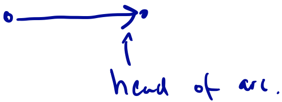
What to do (if anything?) if we have an exp'l # of facets.

will motivate Ellipsoid

r-Arborescences

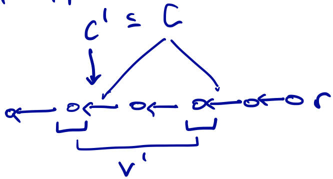
Given Digraph $D = (V, A)$, root r

<u>Def'n</u>: A set of arcs $A' \subseteq A$ is called an r-arborescence if arcs in A' form a spanning tree of D, i.e. every node $\neq r$ is the head of exactly one arc in A'.



<u>Def'n</u>: A subset $C \subseteq A$ is called an r-cut if $C = \delta^-(V')$ for some $V' \subseteq V \setminus \{r\}$.

Minimal cut: defined inclusion-wise. Ex:



Shortest r-Arborescence

If each edge/arc has a length c_{ij}, then for r-arborescence $A' \subseteq A$

$$c(A) = \sum_{(i,j) \in A} c_{ij} .$$

make LP: $B \subseteq A$, $\chi^B \in \{0,1\}^{|A|}$ given by

$$\chi^B_a = \begin{cases} 1 & \text{if } a \in B \\ 0 & a \notin B \end{cases}$$

Each corner of $[0,1]^A$ corresponds to some subset of A.

Arborescences correspond to a subset of the corners of $[0,1]^A$.

$$\boxed{\mathfrak{X} = \text{Conv.hull}\{\chi^B : B \text{ an r-arb}\}}$$

min: $c^T x$
st: $x \in \mathfrak{X}$

// LP where sol'n is shortest r-arb.

r-Cuts and Blocking Relationship — 1/2

Recall: $C \subseteq A$ is an r-cut if

$C = \delta^-(V')$, $V' \subseteq V \setminus \{r\}$.

C is minimal if it is inclusion-wise minimal.

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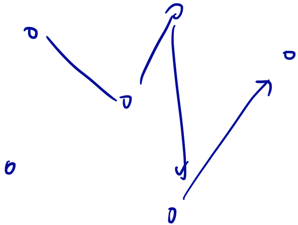
r-Arbor.

$A' \subseteq A$, s.t. every node in $V \setminus r$ is the head of exactly one edge.

r-Cuts and Blocking Relationship — 2/2

<u>Thm</u>: r - Arborescences are exactly the inclusion-wise minimal sets intersecting all r-cuts. Minimal r-cuts are the minimal sets intersecting all r-arborescences.



An Integer Programming Formulation

$$\min: \ c^T x$$
$$\text{st}: \ x \in \mathcal{X} = \text{conv hull}\,\{x^B : B \ \ r\text{-arb}\}$$

r-cut

$$\min: \ c^T x$$
$$\text{st}: \ x(\delta^-(W)) \geq 1, \quad \forall\, \emptyset \neq W \subseteq V \setminus \{r\}$$
$$x_a \in \{0,1\} \quad \forall\, a \in A$$

This IP solves ~~is~~ the min r-Arb. problem.

LP Relaxation is Exact

Thm: [Edmonds, 1967] The LP relaxation of the above IP is exact.

IP feasible set: x : $x(\delta^-(W)) \geq 1$, $\forall\ \emptyset \neq W \subseteq V \setminus \{r\}$

$x_a \in \{0,1\}$

LP x : $x(\delta^-(W)) \geq 1$ " "

$x_a \in [0,1]$

Conv hull of these points =

Exponential Facets

We have formulated the shortest r-Arborescence problem as LP

$$= \min: c^T x$$
$$\text{st: } Ax \leq b$$

$\#$ rows of A $= m =$ $\#$ subsets of $V \setminus \{r\}$.

Finding Violated Constraints

$x(\delta^-(W)) \geq 1, \quad \forall \emptyset \neq W \subseteq V \setminus \{r\}.$

$0 \leq x_a \leq 1$

Given $y \in \mathbb{Q}^A$: is y feasible?

Checking: $\underline{0 \leq y_a \leq 1}$ $\forall a$ — easy.

$\forall s \in V \setminus \{r\}$, find a minimum capacity r-s cut $C_s \subseteq A$ where I treat y as the capacities by e.g., FF.

<u>Claim</u>: $\underbrace{\min\left\{ y(C_s) : s \in V \setminus \{r\} \right\}}_{|V|-1} = \underbrace{\min\left\{ y(\delta^-(W)) : \emptyset \neq W \subseteq V \setminus \{r\} \right\}}_{2^{|V|}}$

<u>Hence</u>: by computing $|V|-1$ min cuts, we find a minimum capacity r-cut, $\delta^-(W^*)$. If $y(\delta^-(W^*)) \geq 1 \Rightarrow y$ is feasible. OW we have found a violated inequality.

Up Next: Ellipsoid Algorithm

We will see: Ellipsoid alg can solve this kind of problem.

Given ~~a convex set~~ polytope P specified by an oracle that either certifies that a point $x \in P$ or provides an inequality feasible for P but violated by x.

(and given ball that contains P and min vol of P)

Ellipsoid alg finds a feasible point of P.